

Yash Sushil

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SUMMARY

Motivated Computer Science undergraduate with expertise in full-stack development, AI integration, and cloud technologies. Proven ability to design and deploy user-focused applications with seamless payment systems and real-time features. Passionate about solving real-world problems through innovative software solutions.

PROJECTS

Real-Time Workout Form Corrector & Analytics Platform

Jun 2026 – Present

Computer Vision System for Real-Time Exercise Form Correction

React.js, FastAPI, Socket.IO, YOLOv8-Pose, MediaPipe, OpenCV, PyTorch, SQLite

- Developed a dual-device, real-time computer vision web application using React.js, FastAPI, and Socket.IO, synchronizing PC and mobile cameras over a local network with sub-second latency for live streaming.
- Deployed YOLOv8-Pose and MediaPipe with OpenCV and PyTorch to track 17-point skeletal landmarks at 30+ FPS, powering a geometric rule engine that evaluates joint angles across 5 core exercises and flags kinetic errors (e.g., knee caving) with real-time corrective feedback alerts to reduce injury risk.
- Designed an asynchronous analytics sub-agent that processes time-series performance logs and auto-generates Markdown performance report cards persisted to SQLite, enabling progress tracking across all 5 supported exercise types.

CachedRAG: Cost-Optimized Document Intelligence Engine

Jun 2026 – Present

Retrieval-Augmented Generation Platform with Semantic Caching

React.js, FastAPI, Redis VSS, Qdrant, Groq (Llama 3.1), Google Gemini API

- Built a Retrieval-Augmented Generation (RAG) platform using React.js and FastAPI, enabling automated, multi-tenant ingestion and indexing of unstructured PDF and text documents.
- Engineered an in-memory semantic caching layer with Redis VSS and cosine similarity to intercept repetitive queries, cutting query latency by over 90% to under 10ms and reducing redundant LLM API calls, backed by Qdrant (in-memory mode) with sentence-transformer embeddings for low-latency vector retrieval on cache misses.
- Architected a dual-LLM pipeline combining Groq (Llama 3.1) for fast streaming text generation with an asynchronous Gemini 2.5 Flash agent to audit and reduce hallucinations, improving response reliability across multi-tenant document queries.

Privacy-Preserving Healthcare AI using Federated Learning

Jan 2025 – Present

Full-Stack Contributor – Decentralized ML System for Diabetes Prediction

React, Python, gRPC, Flower (flwr), TensorFlow/Keras, UCI Pima Dataset

- Built the React frontend for a federated learning healthcare platform, enabling hospital-simulating clients to visualize decentralized MLP model training on the UCI Pima Indians Diabetes dataset, which achieved 80.2% accuracy without any raw patient data leaving local nodes.
- Led full-stack integration between the frontend and gRPC-based backend, connecting the UI to SuperLink/SuperNode communication endpoints for real-time monitoring of TLS-encrypted, HIPAA/GDPR-compliant weight transmission.

Screened Desire | React, Vite, Tailwind CSS, Django REST Framework, Appwrite, TMDB API, Groq LLM, JWT, SQLite

2025

AI-Powered Movie Recommendation & Trailer Platform

- Built a full-stack movie recommendation platform with JWT-authenticated, SQLite-backed Django REST API and React frontend, analyzing user search patterns to surface personalized suggestions via TMDB API, integrated trailer streaming, and a Groq-powered chatbot.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C/C++, MATLAB

Frontend: React.js, React Native, Vite, Tailwind CSS, HTML, CSS

Backend: Node.js, Express.js, Django, FastAPI

Databases & Cloud: MongoDB, PostgreSQL, SQLite, Redis, Qdrant, AWS, Git

Machine Learning & CV: TensorFlow/Keras, PyTorch, YOLOv8-Pose, MediaPipe, OpenCV, Sentence-Transformers

AI/LLM & APIs: OpenAI API, Groq LLM (Llama 3.1), Gemini API, RAG, Vector Search, Semantic Caching, RESTful APIs

Core Concepts: Data Structures, Algorithms, Full-Stack Development, Computer Vision, RAG, Real-Time Systems

EDUCATION

VIT Bhopal University

Bhopal, India

Bachelor of Technology (B.Tech) in Computer Science and Engineering

August 2023 – Present

- CGPA: 8.44

Navkis Educational Centre

Bangalore, India

CBSE XIIth – Science Stream

2019 – 2023

- Xth: 94% | XIIth: 85%

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified DevOps Professional – Oracle

[\[View Certificate\]](#)

Google IT Support Certificate – Google

[\[View Certificate\]](#)

Deloitte Australia – Data Analytics Job Simulation – Forage

[\[View Certificate\]](#)

Programming in Java – Vityarathi

[\[View Certificate\]](#)

Python Essentials – Vityarathi

[\[View Certificate\]](#)